FREIGHT FORWARDING AND SUPPLY CHAIN

Freight Invoice Reconciliation Workflow

Gap Analysis and Business Requirements Document

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This document is read together with the companion deliverable, the RPA Business Case for Freight Invoice Reconciliation Automation, which builds on the requirements established here.

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1. Executive Summary

I was asked to examine the freight invoice reconciliation process operating inside a CargoWise One freight forwarding environment, because the finance and operations teams suspected that carrier overcharges were passing through unchallenged. I conducted a structured gap analysis across the end to end workflow, sampled a representative set of carrier invoices, and traced how each invoice moved from receipt through to approval and payment.

What I found is best described as a control gap rather than a speed problem. The team does reconcile invoices, but the comparison happens at the invoice total level against the original freight quote, and there is no systematic three way match against the underlying shipment record. When I sampled invoices, only about 15 percent received a full three way comparison, and roughly 6 percent carried a rate variance that no one had detected. Because the checking is manual and inconsistent, genuine discrepancies are caught by chance rather than by design.

The target state I set out in this document keeps the human decision where it adds value, the judgement on whether a discrepancy is real, and removes the human from the repetitive comparison work where accuracy suffers under volume. The metrics below frame the size of the opportunity and anchor the requirements that follow.

Measure	Current State	Target State
Average invoice processing time	22 minutes per invoice	Under 6 minutes per invoice
Invoices receiving a full three way match	15 percent	100 percent
Undetected rate variance rate	6 percent of invoices	Below 1 percent of invoices
Approval cycle time	3.1 days	Under 1 day
Basis of comparison	Invoice total against quote	Line item charge code against quote and shipment record

2. Background and Business Context

2.1 Organisational Context

The organisation is a freight forwarder that books and manages international shipments on behalf of importers and exporters, and operates on the CargoWise One platform for shipment management, accounting, and document control. Carriers, which include shipping lines, airlines, and road hauliers, issue invoices for the services rendered against each shipment. These invoices are the single largest category of direct cost the business carries, so the accuracy of the charges directly affects gross margin on every consignment.

Invoices arrive through several channels and in inconsistent formats. Some carriers send a structured PDF, others send an Excel attachment, and the layout, the charge code naming, and the level of detail differ from one carrier to the next. The finance team receives these invoices, opens the related shipment in CargoWise One, and reconciles the charges before approving the invoice for payment.

2.2 Problem Statement

The reconciliation that takes place today compares the total amount on the carrier invoice against the total on the freight quote that was agreed with the customer or negotiated with the carrier. If the two totals are close, the invoice is approved. This approach has two structural weaknesses. First, it compares totals rather than the individual charge code lines, so an overcharge on one line that is offset by an undercharge or an omission on another can net out and pass undetected. Second, it does not cross check the charges against the shipment record itself, the record that holds the actual container count, weight, route, and accessorial events, so charges for services that were never performed are not systematically challenged.

The result is that the team believes it is reconciling invoices when in practice it is performing a total level sanity check. The work is slow because it is manual, but the more serious issue is that the control it is meant to provide is not actually present for the majority of invoices.

2.3 Business Impact

I quantified the impact across four dimensions, drawing on the sampling exercise and on timing observations taken during the review.

Impact Area	Description	Observed Effect
Financial leakage	Rate variances that are not detected are paid in full	6 percent of invoices carry undetected variance
Control assurance	Most invoices never receive a true three way match	Only 15 percent are fully reconciled
Processing effort	Manual comparison across mismatched formats	22 minutes of analyst time per invoice

Impact Area	Description	Observed Effect
Cash and supplier cycle	Slow approval delays payment and dispute raising	3.1 day average approval cycle

3. Scope

3.1 In Scope

1. The reconciliation of inbound carrier invoices received in PDF and Excel formats against the corresponding freight quote and shipment record held in CargoWise One.
2. The comparison of invoice charges at the individual charge code line level rather than at the invoice total level.
3. The establishment of a three way match between the carrier invoice, the freight quote, and the shipment record.
4. The definition of a configurable variance tolerance and an escalation path for charges that fall outside that tolerance.
5. The functional and non functional requirements that a future automation or system change must satisfy to close the identified gaps.

3.2 Out of Scope

1. Changes to the carrier rate negotiation process or to the commercial terms agreed with carriers.
2. Reconciliation of customer billing or accounts receivable, which is a separate workflow.
3. Replacement or reconfiguration of the CargoWise One platform itself beyond the data access required for reconciliation.
4. The selection of a specific automation vendor, which is addressed in the companion RPA Business Case.

4. Stakeholders

I identified the following stakeholders during the review and confirmed their interest and influence in conversation with the operations lead.

Stakeholder	Role in the Process	Primary Interest
Finance Manager	Owns invoice approval and payment	Accurate charges and a defensible audit trail
Accounts Payable Analyst	Performs the day to day reconciliation	A faster and clearer comparison method
Operations Manager	Owns shipment data quality	Confidence that charges reflect services performed
Customer Service Coordinator	Manages customer freight quotes	Quotes that match the charges ultimately invoiced
Internal Audit and Compliance	Reviews financial controls	Evidence that a genuine three way control exists

5. As Is Process Analysis

5.1 Current Workflow

I walked the current process step by step with the accounts payable analyst and recorded the workflow exactly as it runs today, including the points where judgement is applied and where steps are skipped under time pressure.

Step	Activity	Performed By	Observation
1	Carrier invoice received by email in PDF or Excel	Carrier	Format and layout vary by carrier
2	Invoice manually read and key charges noted	AP Analyst	Re keying introduces transcription risk
3	Related shipment opened in CargoWise One	AP Analyst	Shipment record not used for line checks
4	Invoice total compared to freight quote total	AP Analyst	Comparison is at total level only
5	If totals are close, invoice approved	AP Analyst	Offsetting line errors pass undetected
6	If totals differ materially, query raised	AP Analyst	Trigger is subjective, not threshold based
7	Approved invoice posted for payment	Finance Manager	Average cycle reaches 3.1 days

5.2 Root Cause Analysis

To avoid designing requirements that treat symptoms, I applied the five why method to the two most material problems I observed. The analysis is written below in prose so that the chain of reasoning is explicit.

Root cause one, undetected rate variance

I started from the observation that 6 percent of invoices carried a variance that no one had caught. I asked why the variance was not caught, and the answer was that the analyst only compared invoice totals. I asked why only totals were compared, and the answer was that comparing every charge code line by hand across mismatched formats was too slow to sustain at the volume received. I asked why the comparison was so slow, and the answer was that invoice data arrived in inconsistent PDF and Excel layouts that had to be read and re keyed manually. I asked why that inconsistency was tolerated, and the answer was that there was no extraction step to normalise invoice data into a common structure. I asked why no normalisation existed, and the answer was that the process had grown up as a manual task and had never been redesigned around the shipment record as the source of truth. The true root cause is therefore the absence of a structured, line level comparison anchored to the shipment record, not a lack of diligence on the part of the analyst.

Root cause two, slow approval cycle

I started from the 3.1 day average approval cycle. I asked why approval took so long, and the answer was that each invoice consumed about 22 minutes of manual effort and queued behind others. I asked why each invoice took so long, and the answer was that the analyst manually located the shipment, read the invoice, and reconciled the figures by hand. I asked why this could not be done faster, and the answer was that there was no systematic link between the invoice and the shipment record that could surface the relevant comparison automatically. I asked why that link was missing, and the answer was that reconciliation relied on the analyst's memory and manual navigation rather than on a defined matching rule. I asked why no matching rule existed, and the answer was that the process had never been specified to the level of charge codes and tolerances. The true root cause is the lack of a defined matching rule and tolerance, which forces every invoice through a slow manual path regardless of whether it is straightforward or genuinely exceptional.

6. Gap Analysis

The table below sets out the three gaps I identified, each expressed as the difference between the current state and the target state, with the business impact and the priority I assigned.

Gap Area	As Is State	To Be State	Impact	Priority
Three way matching	Invoice total compared to quote only, no shipment cross check	Each charge code line matched against quote and shipment record	Closes the financial control gap behind the 6 percent variance	**High**
Variance tolerance and escalation	Subjective decision on when totals differ enough to query	Configurable tolerance threshold with automatic escalation of genuine exceptions	Makes the control consistent and auditable	**High**
Invoice data extraction consistency	PDF and Excel read and re keyed by hand	Invoice data extracted and normalised into a common structure	Removes transcription risk and the 22 minute manual effort	**Medium**

7. Functional Requirements

I grouped the functional requirements under the three gap areas so that every requirement is traceable to a gap and ultimately to a root cause. Each requirement carries a unique identifier, a priority, and an acceptance criterion that defines how I would confirm it has been met.

7.1 Three Way Matching

ID	Requirement	Priority	Acceptance Criteria
FR-001	The solution shall match each carrier invoice charge code line against the corresponding line on the freight quote.	High	Every invoice line is paired to a quote line or flagged as unmatched
FR-002	The solution shall cross check each charge against the shipment record in CargoWise One for the services actually performed.	High	Charges with no supporting shipment event are flagged for review
FR-003	The solution shall record the outcome of the three way match against each invoice for audit purposes.	High	A stored match result is retrievable for 100 percent of invoices

7.2 Variance Tolerance and Escalation

ID	Requirement	Priority	Acceptance Criteria
FR-004	The solution shall apply a configurable variance tolerance, expressed as both a percentage and an absolute amount, to each matched line.	High	An authorised user can change the tolerance without code change
FR-005	The solution shall automatically approve invoices where every line falls within tolerance.	High	In tolerance invoices proceed without manual intervention
FR-006	The solution shall escalate to a human approver only those invoices that contain a line outside tolerance, with the variance detail attached.	High	Escalated cases show the specific line, expected value, and actual value

7.3 Invoice Data Extraction Consistency

ID	Requirement	Priority	Acceptance Criteria
FR-007	The solution shall extract charge data from carrier invoices supplied in PDF and Excel formats.	Medium	Both formats are processed without manual re keying

ID	Requirement	Priority	Acceptance Criteria
FR-008	The solution shall normalise extracted charges into a common charge code structure for comparison.	Medium	Carrier specific labels are mapped to standard charge codes
FR-009	The solution shall route invoices it cannot read with confidence to a manual queue rather than guessing.	Medium	Low confidence extractions are held, not auto approved

8. To Be Process Design

The redesigned workflow below keeps the analyst in control of genuine exceptions while removing them from the repetitive comparison. The final column states plainly what changed relative to the current process.

Step	Activity	Performed By	Change from As Is
1	Carrier invoice received and data extracted automatically	Automation	Replaces manual reading and re keying
2	Charges normalised to standard charge codes	Automation	New step, removes format inconsistency
3	Each line matched against quote and shipment record	Automation	Replaces total level comparison
4	Variance tolerance applied to every line	Automation	New step, makes the control consistent
5	In tolerance invoices approved automatically	Automation	Removes manual effort on clean invoices
6	Out of tolerance invoices escalated with detail	Automation to AP Analyst	Analyst now sees only genuine exceptions
7	Exception reviewed and resolved	AP Analyst	Judgement focused on real discrepancies
8	Approved invoice posted for payment	Finance Manager	Cycle time reduced from 3.1 days

9. Non Functional Requirements

ID	Category	Requirement
NFR-001	Performance	The solution shall process a standard invoice end to end in under 6 minutes
NFR-002	Accuracy	Charge extraction shall achieve at least 98 percent field level accuracy on established carrier formats
NFR-003	Auditability	Every match decision and tolerance setting shall be logged with a timestamp and user reference
NFR-004	Configurability	Tolerance thresholds and charge code mappings shall be maintainable by business users
NFR-005	Security	Access to invoice and shipment data shall follow the existing CargoWise One permission model
NFR-006	Reliability	The solution shall fail safe by holding uncertain invoices rather than auto approving them

10. User Acceptance Testing Approach

I propose that user acceptance testing confirms the control behaves correctly at the line level and that the escalation path surfaces only genuine exceptions. The testing should be run by the accounts payable analyst and the finance manager using a curated set of real historical invoices that includes known clean cases, known overcharges, and deliberately awkward formats. The scenarios below define the minimum coverage I would sign off against.

ID	Test Scenario	Expected Result
UAT-01	A clean invoice where every line matches quote and shipment	Auto approved with a stored match result
UAT-02	An invoice with one line above the variance tolerance	Escalated with the specific line and variance shown
UAT-03	An invoice with an offsetting overcharge and undercharge	Both lines flagged despite the total appearing correct
UAT-04	A charge with no supporting event in the shipment record	Flagged as unsupported and held for review
UAT-05	An invoice in an unfamiliar carrier format	Held in the manual queue rather than auto approved

11. Implementation Plan

I structured delivery into four phases so that the control logic is proven on a small set of carriers before it is widened. The durations are indicative and assume the requirements in this document are accepted.

Phase	Key Activities	Owner	Duration
1. Design and mapping	Confirm charge code mappings and tolerance rules with finance	Business Analyst	2 weeks
2. Build	Configure extraction, matching, and escalation logic	Automation Developer	4 weeks
3. Test and UAT	Run the UAT scenarios against historical invoices	AP Analyst and Finance Manager	2 weeks
4. Deploy and stabilise	Go live on priority carriers and monitor exceptions	Operations Manager	2 weeks

12. Success Metrics

I will judge the change a success against the baseline I measured during the review, assessed three months after go live.

Metric	Baseline	Three Month Target
Invoices receiving a full three way match	15 percent	100 percent
Undetected rate variance	6 percent of invoices	Below 1 percent
Average processing time per invoice	22 minutes	Under 6 minutes
Approval cycle time	3.1 days	Under 1 day

13. Assumptions and Constraints

13.1 Assumptions

1. The freight quote held in CargoWise One is the agreed reference rate and is reliable enough to match against.
2. The shipment record accurately reflects the services performed, so it can serve as the third leg of the match.
3. Carrier invoice formats, while inconsistent between carriers, are stable enough per carrier to be mapped.
4. Finance is willing to own and maintain the variance tolerance settings after go live.

13.2 Constraints

1. The solution must operate within the existing CargoWise One environment and its permission model.
2. No change to carrier commercial terms can be assumed as part of this work.
3. Historical invoice data available for testing is limited to the carriers currently in active use.
4. Delivery effort must fit within the indicative phase durations set out in the implementation plan.

14. Document Control

Version	Date	Author	Summary of Change
0.1	18 April 2026	N M Ziyaf	Initial draft following process walkthrough
0.2	24 April 2026	N M Ziyaf	Added gap analysis and functional requirements
1.0	30 April 2026	N M Ziyaf	Finalised for stakeholder review